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EXPERIMENT 12

Perspective Transformation on an Image

AIM

To perform a perspective transformation on the given image.

PROGRAM

```
import cv2
import numpy as np

img = cv2.imread("image1.jpg")
rows, cols = img.shape[:2]

# A perspective transform is defined by four corresponding points
src_points = np.float32([[0, 0], [cols - 1, 0],
                         [0, rows - 1], [cols - 1, rows - 1]])
dst_points = np.float32([[0, 0], [cols - 1, 0],
                         [int(0.33 * cols), rows - 1], [int(0.66 * cols), rows - 1]])

M = cv2.getPerspectiveTransform(src_points, dst_points)
perspective_img = cv2.warpPerspective(img, M, (cols, rows))

cv2.imwrite("perspective_transformed.jpg", perspective_img)
print("Perspective matrix:\n", M)

cv2.imshow("Original", img)
cv2.imshow("Perspective Transformed", perspective_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT

image1.jpg (input image)

OUTPUT

perspective_transformed.jpg (the bottom edge is narrowed, producing a depth effect)

RESULT

Thus a perspective transformation was successfully applied to the given image using cv2.getPerspectiveTransform() and cv2.warpPerspective().